Memory map and peripheral contract

Five-Stage Pipelined Microcontroller for a Washing Machine
Logisim-Evolution 5.0.0 • Assignment 2

All values below are hexadecimal unless stated. Data address space is byte-addressed; instruction ROM is separate and word-addressed. Peripheral writes execute once at the rising edge of a valid MEM-stage ST.

Address	Access / reset	Meaning
00-7F	RW / unspecified	128-byte RAM; contents not cleared by reset
80-EF	R=00; W ignored	Reserved; cannot alias RAM or I/O
F0	R / live	bit0 START; bit1 DOOR_CLOSED; bits3:2 MODE; bits7:4=0
F1	R / live	Unsigned WATER_LEVEL[7:0]
F2	R / live	TEMP[7:0]
F3	R / live	DRUM_RPM[7:0]
F4	RW / 00	Actuator requests: 0 VALVE, 1 HEATER, 2 MOTOR_EN, 3 DIR, 4 PUMP, 5 LOCK, 6 BUZZER; bit7 ignored/read0
F5	RW / 00	DISPLAY[7:0]
F6	R / 00	TIMER_COUNT[7:0]
F7	RW / 00	TIMER_LIMIT[7:0]; 00 disables comparison
F8	RW / 00	bit0 ENABLE; bit1 write-1-clear COUNT/MATCH; bit1 reads0; other bits ignored/read0
F9	R / 00	bit0 sticky MATCH; other bits0
FA	W / R=00	bit0=1 requests START acceptance/safe rearm; other bits ignored
FB	R / 00	Accepted MODE_SNAPSHOT in bits1:0; others0
FC	R / 00	bit0 last ACK success; bit1 TRIP; others0
FD-FF	R=00; W ignored	Unimplemented peripheral addresses

Timer: write F7=nonzero limit; write F8=03 to enable and clear. The clear edge does not increment. Each following enabled edge increments modulo 256; if the new count equals the old nonzero limit, MATCH sets and stays set. F8=00 disables without clearing. F8=02 disables and clears. F8 enable updates on the write edge; increment uses the old enable unless a clear is present.

Input sampling: valid LD captures current read data at the MEM edge. Hold sensor pins steady through that edge. Read-only writes are ignored. Reset asynchronously clears actuator requests, DISPLAY, all timer state, accepted mode, pending START, ACK status and TRIP.

Safety/readback: F4 reports Processor requests; it is not the physical motor state. Observe ACTUAL_MOTOR separately. Successful FA acceptance requires a fresh qualified START edge, closed door and motor request=0. Command examples: 21 lock+fill, 24/2C wash directions, 30 lock+pump, 20 lock only, 40 buzzer with motor stopped and door unlocked.